higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

CIVIL AND STRUCTURAL STEEL WORK DETAILING

(First Paper)

NQF LEVEL 4

(12050674)

1 March 2019 (Y-Paper)
13:00–15:00

REQUIREMENTS: **A3 drawing sheets**
Standard hot-rolled structural steel section table (BOE 8/1)

Calculators and drawing instruments may be used.

This question paper consists of 5 pages, 1 formula sheet and 1 addendum.

NCV

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TIME: 2 HOURS
MARKS: 75

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each question on a NEW page.
5. Rule off across the page on completion of each question.
6. Make drawings and sketches clear enough to show detail.
7. Fully dimension and label ALL drawings.
8. Write neatly and legibly.

NCV

QUESTION 1: CALCULATIONS FOR CIVIL AND STRUCTURAL STEEL WORK

1.1 Calculate the amount of steel required for a column with a width of 500 mm and a height of 500 mm if it is subjected to an un-factored load of 3 000 kN. The strength of the concrete is 25 MPa and of the steel 350 MPa. (7)

1.2 A 60 kN tensile force is applied to a 60 × 60 × 10 back-to-back angle section. The yield strength of steel is 450 MPa.

Calculate the tensile resistance. (3)

1.3 Determine the maximum compressive resistance C_r that a section 254 × 254 × 73, f_y = 350 MPa will be able to resist. Use a K value of 1. The slenderness ratio is 1,6. L = 6 000 mm and E = 200 GPa. (3)

1.4 Determine the maximum compressive resistance C_r that a column section 254 × 254 × 73, f_y = 350 MPa will be able to resist. Use a K value of 2. L = 6 000 mm and E = 200 GPa.

Refer to the ADDENDUM (attached). (5)

1.5 FIGURE 1 shows a loaded simply supported beam.

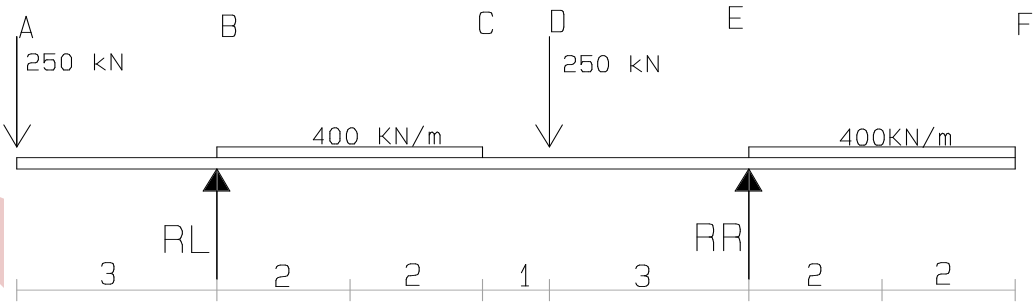


FIGURE 1

1.5.1 Calculate the reactions. (8)

1.5.2 Calculate the maximum shear force. Show ALL calculations. (6)

[32]

QUESTION 2: INTRODUCTION TO CIVIL AND STRUCTURAL STEEL WORK

2.1 Name THREE apparatuses used for a slump test. (3)

2.2 Name TWO supports associated with beams and give the advantage of each. (2 × 2) (4)

2.3 Explain the procedure for the analysis of steel beams. (7)

2.4 2.4.1 Explain each of the following steel-type abbreviations:

(a) R

(b) Y (2)

2.4.2 Explain each of the following location abbreviations:

(a) T

(b) B

(c) EW

(d) ABR

(e) STG

(f) ALT

(g) TOG (7)

2.5 Give a concise definition of a *pile cap*. (2)
[25]

NCM

QUESTION 3: STRUCTURAL STEEL WORK DETAILING AND SPECIFICATIONS

Choose a description from COLUMN B that matches an item in COLUMN A. Write only the letter (A–H) next to the question number (3.1–3.8) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1	KL/r	A	maximum compressive force
3.2	Triangle of forces	B	distribution of forces that works over an entire area
3.3	N/m^2	C	equivalent of gravitational force
3.4	M_{MAX}/h	D	shear force
3.5	Pascal	E	pushing or pulling action
3.6	F = ma	F	another name used to refer to vector addition
3.7	F_S	G	slenderness ratio
3.8	Weight	H	unit for stress

(8 × 1) **[8]**

QUESTION 4: PROPERTIES OF STRUCTURAL STEEL WORK AND REINFORCED CONCRETE

4.1 Differentiate between a *cube test* and a *slump test*. (2 + 2) (4)

4.2 With the aid of sketches, state the properties of a *shear slump* and a *collapsed slump*. (3 + 3) (6)

[10]

TOTAL: 75

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FORMULA SHEET

$F = ma$

$W = mg$

$M_B = F_A d_{AB}$

$\sum Fy = 0$

$\sum Fx = 0$

$\sum M = 0$

$\sigma = \frac{F}{A}$

$F_C = F_S$

$M = F_S \times$ perpendicular distance

$M = F_C \times$ perpendicular distance

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$UDL = Y_{CON} \times$ cross-sectional area

$F_C = 0{,}45\sigma cb(0{,}9x)$

$F_C = 0{,}45\sigma bh$

$F_S = 0{,}87\sigma y A_{st}$

$F_S = 0{,}73\sigma y A_{sc}$

$N = F_C + F_S$

$Q_B A - N - Y_S(1)(A) - Y_C(h)(A) = 0$

$A = \frac{N}{QB - YS(1) - YC(h)}$

$Q_U = \frac{UDL}{A}$

$M_r = \emptyset Z_{Pl} f_y$

$M_r = \emptyset Z_e f_y$

Self-weight of a beam $= m \times a$

$\lambda = \frac{KL}{r}\sqrt{\frac{fy}{\pi^2 E}}$

$C_r = \emptyset A f_y (1 + \lambda^{2,68})^{-1/1,34}$

$T_r = \emptyset A fy$

$F_{AE} \cos\theta + F_{AC} = 0$

$F_{AE} = -\frac{100}{Sin30}$

$F_{AB} \sin\theta - F_{BE} = 0$

Maximum compressive force $= M_{MAX}/h$

Maximum tensile force $= F_{SMAX}/\cos\emptyset$

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ADDENDUM

Table of properties of H-sections

H-section	m (kg/m)	h (mm)	b (mm)	t_w (mm)	t_f (mm)	h_w (mm)	A ($10^3 mm^2$)	Z_e ($10^3 mm^2$)	Z_{pl} ($10^3 mm$)	r (mm)
254 × 254 × 73	73,1	254,2	254,0	8,6	14,2	200	9,29	896	990	64,6
254 × 254 × 89	88,9	250,4	255,9	10,5	17,3	200	11,4	1 100	1 230	65,2
254 × 254 × 107	107	266,7	258,3	13	20,5	200	13,7	1 310	1 480	65,7
254 × 254 × 132	132	276,4	261,0	15,6	25,1	201	16,8	1 620	1 860	66,6
254 × 254 × 167	167	289,1	264,5	19,2	31,7	200	212,2	2 070	2 420	67,9
305 × 305 × 97	96,9	307,8	304,8	9,9	15,4	247	12,3	1 440	1 590	76,8
305 × 305 × 118	118	314,5	306,8	11,9	18,7	247	15,0	1 760	1 950	77,6
305 × 305 × 137	137	320,5	308,7	13,8	21,7	247	17,4	2 040	2 290	78,2
305 × 305 × 158	158	327,2	310,6	15,7	25	247	20,1	2 360	2 670	78,9

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